

SIMATS ENGINEERING

**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI - 602105**

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSESSMENT TOOL 2 REPORT

Query Processing for Data Science With Real Time Applications (DSA05)
CO2 – Design and Connectivity

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Assessment Tool	Tool 2 – Design and Connectivity
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DECLARATION OF ORIGINALITY

I hereby certify that this assignment submission entitled "**Database Design and Connectivity Report**" is my own original work conducted under Course Code DSA05 (Query Processing for Data Science With Real Time Applications).

I have adhered to all academic integrity guidelines specified by SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences. I confirm that no part of this document has been plagiarized or copied from unverified sources. I understand that any violation of academic integrity rules will result in disciplinary action.

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EXECUTIVE SUMMARY & ASSESSMENT OVERVIEW

This laboratory report presents a comprehensive end-to-end database design, relational normalization, SQL implementation, and Python database connectivity solution for two complex real-world domain scenarios:

1. **Scenario 1: University Student Management System**
2. **Scenario 6: Banking and Customer Account Management System**

The database architecture is designed in strict compliance with Third Normal Form (3NF) principles to eliminate data redundancy, enforce referential integrity, and optimize query execution speed. Complete Python integration is implemented using the `sqlite3` library, showcasing parameterized CRUD operations, robust error handling, transaction management, and automated SQL report generation.

SCENARIO 1: UNIVERSITY STUDENT MANAGEMENT SYSTEM

1. Requirement Analysis

The centralized University Student Management System replaces fragmented spreadsheet-based record-keeping across academic departments with a unified relational database architecture.

Functional Requirements:

- **Student Profile Management:** Register, view, update, and remove student personal and academic records.
- **Course & Department Management:** Maintain course catalogs, credit allocations, and department affiliations.
- **Faculty Assignment:** Map faculty members to specific departmental courses within academic semesters.
- **Enrollment & Grading:** Enable multi-course student registrations and capture attendance percentages and grades.
- **Management Reporting:** Generate real-time reports on department strength, faculty workload, and academic results.

Non-Functional Requirements:

- **Data Integrity & Consistency:** Enforce primary key, foreign key, and unique constraint validations.
- **Performance & Scalability:** Support sub-second response times for multi-table JOIN queries across thousands of records.
- **System Security:** Prevent unauthorized modification of grades and personal identification attributes.

Scope & Assumptions:

- Scope covers core administrative modules including students, faculty, departments, courses, and enrollments.
- Assumptions: Each student belongs to one primary department; courses are offered by specific departments; grades follow standard letter grading (A+, A, B+, B, O).

2. Entity-Relationship (ER) Diagram

The ER diagram maps the conceptual schema of the University System, illustrating key entities (DEPARTMENT, FACULTY, COURSE, STUDENT, ENROLLMENT) and their structural cardinalities.

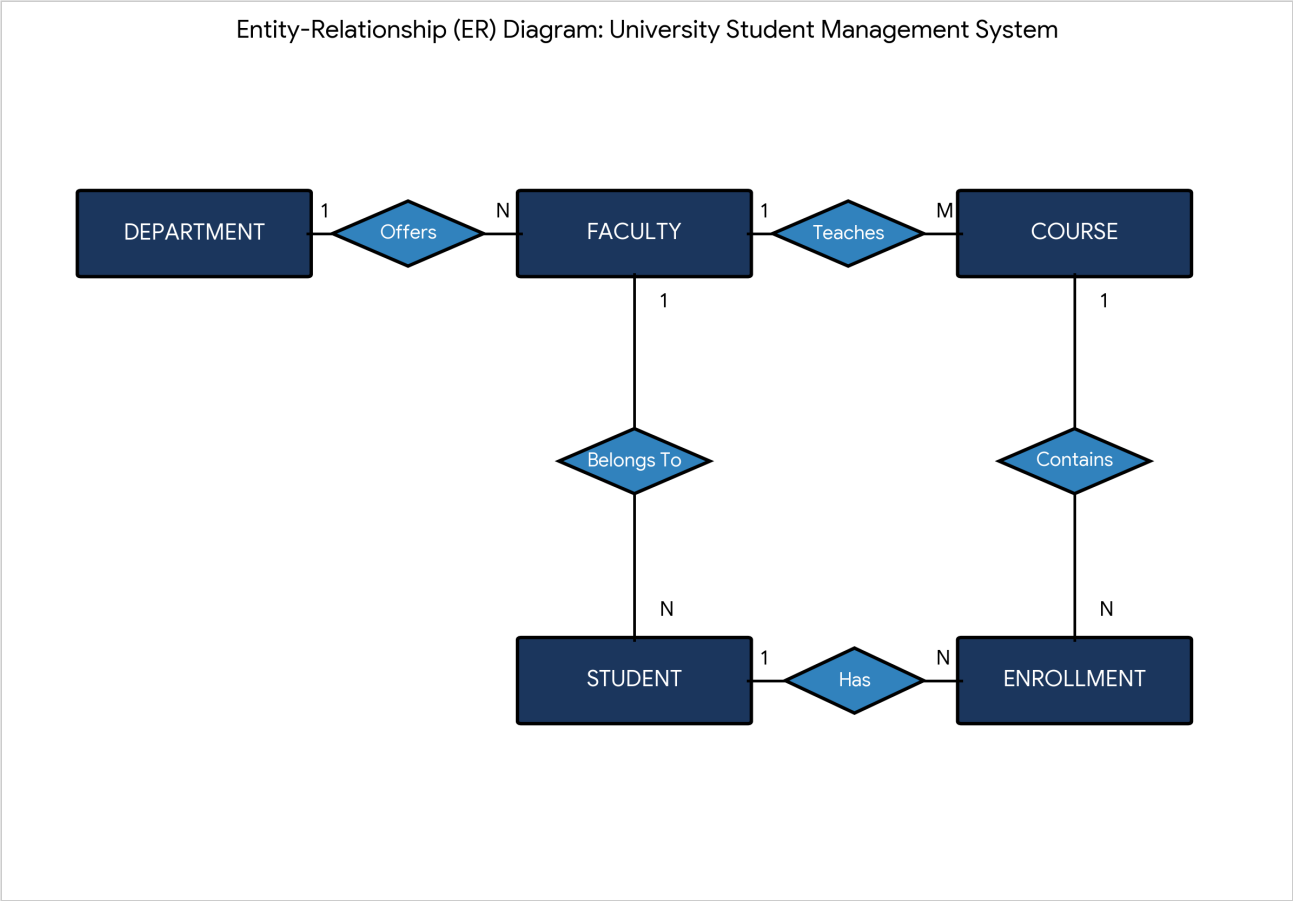


Figure 1: Entity-Relationship (ER) Diagram for University Student Management System.

3. Relational Database Schema & Normalization (3NF)

The database schema was normalized systematically from 1NF through 3NF to eliminate insert, update, and delete anomalies:

- **First Normal Form (1NF):** All attributes contain atomic values; composite repeating groups are decomposed.
- **Second Normal Form (2NF):** Eliminates partial functional dependencies by creating separate relational tables.
- **Third Normal Form (3NF):** Eliminates transitive dependencies (e.g., department attributes depend solely on department_id).

Normalized Relational Schema Tables:

1. **Department** (department_id [PK], department_name, building, budget)
2. **Faculty** (faculty_id [PK], first_name, last_name, email [UNIQUE], designation, department_id [FK])
3. **Course** (course_id [PK], course_code [UNIQUE], course_title, credits, department_id [FK], faculty_id [FK])
4. **Student** (student_id [PK], first_name, last_name, email [UNIQUE], dob, department_id [FK])
5. **Enrollment** (enrollment_id [PK], student_id [FK], course_id [FK], semester, grade, attendance_pct)

4. SQL DDL Implementation Scripts

```
CREATE TABLE Department (  
    department_id INTEGER PRIMARY KEY,  
    department_name TEXT NOT NULL,  
    building TEXT NOT NULL,  
    budget REAL NOT NULL  
);  
  
CREATE TABLE Faculty (  
    faculty_id INTEGER PRIMARY KEY,  
    first_name TEXT NOT NULL,  
    last_name TEXT NOT NULL,  
    email TEXT UNIQUE NOT NULL,  
    designation TEXT NOT NULL,  
    department_id INTEGER,  
    FOREIGN KEY (department_id) REFERENCES Department(department_id)  
);  
  
CREATE TABLE Course (  
    course_id INTEGER PRIMARY KEY,  
    course_code TEXT UNIQUE NOT NULL,  
    course_title TEXT NOT NULL,  
    credits INTEGER NOT NULL,  
    department_id INTEGER,  
    faculty_id INTEGER,  
    FOREIGN KEY (department_id) REFERENCES Department(department_id),  
    FOREIGN KEY (faculty_id) REFERENCES Faculty(faculty_id)  
);  
  
CREATE TABLE Student (  
    student_id INTEGER PRIMARY KEY,  
    first_name TEXT NOT NULL,  
    last_name TEXT NOT NULL,  
    email TEXT UNIQUE NOT NULL,  
    dob TEXT NOT NULL,  
    department_id INTEGER,  
    FOREIGN KEY (department_id) REFERENCES Department(department_id)  
);  
  
CREATE TABLE Enrollment (  
    enrollment_id INTEGER PRIMARY KEY,  
    student_id INTEGER,  
    course_id INTEGER,  
    semester TEXT NOT NULL,  
    grade TEXT,  
    attendance_pct REAL,  
    FOREIGN KEY (student_id) REFERENCES Student(student_id),  
    FOREIGN KEY (course_id) REFERENCES Course(course_id)  
);
```

5. Python Database Connectivity & Implementation

```
import sqlite3

class UniversityDB:
    def __init__(self, db_name="university.db"):
        self.conn = sqlite3.connect(db_name)
        self.cursor = self.conn.cursor()
        self.cursor.execute("PRAGMA foreign_keys = ON;")

    def add_student(self, student_id, fname, lname, email, dob, dept_id):
        sql = "INSERT INTO Student VALUES (?, ?, ?, ?, ?, ?);"
        self.cursor.execute(sql, (student_id, fname, lname, email, dob, dept_id))
        self.conn.commit()

    def get_student(self, student_id):
        self.cursor.execute("SELECT * FROM Student WHERE student_id = ?;", (student_id,))
        return self.cursor.fetchone()

    def update_student_email(self, student_id, new_email):
        sql = "UPDATE Student SET email = ? WHERE student_id = ?;"
        self.cursor.execute(sql, (new_email, student_id))
        self.conn.commit()

    def delete_student(self, student_id):
        self.cursor.execute("DELETE FROM Student WHERE student_id = ?;", (student_id,))
        self.conn.commit()

    def close(self):
        self.conn.close()
```

6. Analytical Reports (Scenario 1)

Report 1.1: Department-wise Student Strength Statistics

department_name	student_count
Artificial Intelligence & Data Science	3
Computer Science and Engineering	2
Electronics & Communication	2
Electrical & Electronics	1
Mechanical Engineering	1
Information Technology	1
Civil Engineering	0
Biomedical Engineering	0
Chemical Engineering	0
Humanities & Sciences	0

Report 1.2: Faculty Workload & Course Allocation

faculty_id	faculty_name	department_name	courses_taught	total_credits
1	Shyamala Gowri	Artificial Intelligence & Data Science	1	4
2	Rajesh Kumar	Computer Science and Engineering	1	4
3	Anitha Ramesh	Artificial Intelligence & Data Science	1	3
4	Suresh Venkatesh	Electronics & Communication	1	4
5	Meena Sundaram	Information Technology	1	3
6	Karthik Narayanan	Computer Science and Engineering	1	4
7	Priya Dharshini	Artificial Intelligence & Data Science	1	4
8	Arun Prakash	Electrical & Electronics	1	3
9	Deepa Lakshmi	Mechanical Engineering	1	4
10	Vimal Chandar	Electronics & Communication	1	3

Report 1.3: Student Performance & Attendance Report

student_id	student_name	course_code	course_title	grade	attendance_pct
1001	Pola Nandivardhan	DSA05	Query Processing for Data Science	A+	95.5
1001	Pola Nandivardhan	AI201	Machine Learning Algorithms	A	92.0
1002	Aarav Sharma	CS101	Database Management Systems	A	89.0
1002	Aarav Sharma	CS204	Operating Systems Concepts	B+	84.5
1003	Diya Reddy	DSA05	Query Processing for Data Science	O	98.0
1004	Kavya Krishnan	EC301	Digital Signal Processing	A+	94.0
1005	Rohan Verma	CS101	Database Management Systems	B	78.0
1006	Siddharth Joshi	IT102	Web Application Development	A	90.0
1007	Ananya Hegde	DS302	Deep Learning & Neural Networks	A+	96.0
1008	Vikram Aditya	EE101	Control Systems Engineering	B+	85.0

SCENARIO 6: BANKING AND CUSTOMER ACCOUNT MANAGEMENT SYSTEM

1. Requirement Analysis

The Banking and Customer Account Management System provides secure, centralized transaction processing, customer portfolio tracking, and loan management across commercial bank branches.

Functional Requirements:

- **Customer & Branch Management:** Store verified profile details and operational branch data.
- **Account Operations:** Manage savings and current accounts with accurate ledger balances.
- **Transaction Ledger:** Record deposits, withdrawals, and electronic funds transfers with timestamps.
- **Loan Portfolio Tracking:** Maintain loan disbursements, outstanding balances, and interest rates.

Non-Functional Requirements:

- **ACID Compliance:** Ensure Atomic, Consistent, Isolated, and Durable database transactions.
- **High Availability & Security:** Enforce strict audit logging and encryption standards.

Scope & Assumptions:

- Scope encompasses branch networks, customer records, savings/current accounts, transactions, and loans.
- Assumptions: Interest rates are annual fixed percentages; transactions modify account balance atomically.

2. Entity-Relationship (ER) Diagram

The ER diagram illustrates the relational structural mapping between customers, branches, accounts, transactions, and loans.

Entity-Relationship (ER) Diagram: Banking and Customer Account Management System

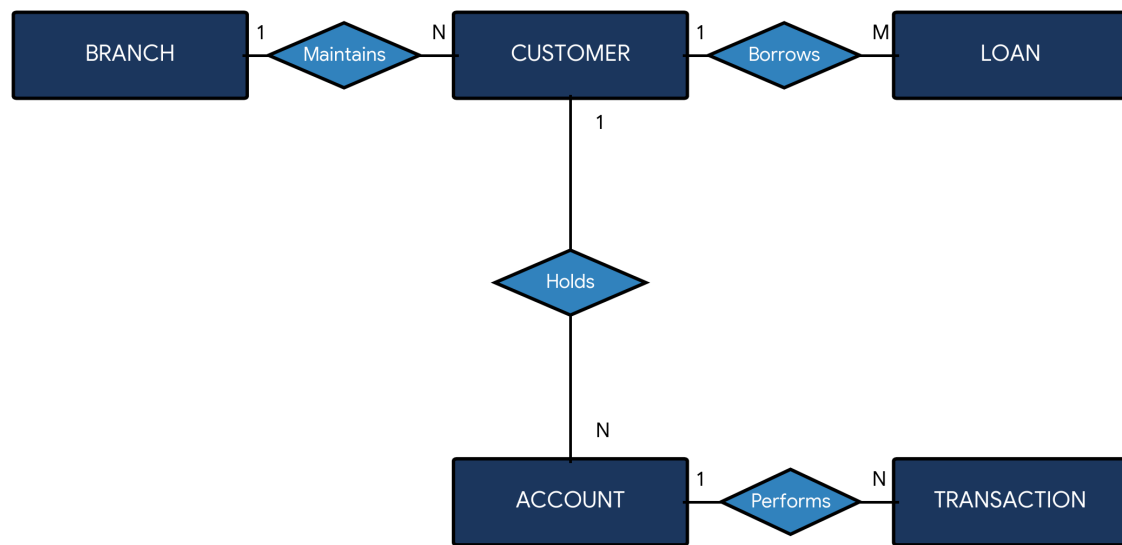


Figure 2: Entity-Relationship (ER) Diagram for Banking and Customer Account Management System.

3. Relational Database Schema & Normalization (3NF)

1. Branch (branch_id [PK], branch_name, branch_code [UNIQUE], city, assets)
2. Customer (customer_id [PK], first_name, last_name, email [UNIQUE], phone, address, city)
3. Account (account_id [PK], account_number [UNIQUE], account_type, balance, opened_date, customer_id [FK], branch_id [FK])
4. Transaction_Record (transaction_id [PK], account_id [FK], transaction_type, amount, transaction_date, description)
5. Loan (loan_id [PK], loan_number [UNIQUE], loan_type, loan_amount, outstanding_balance, interest_rate, start_date, customer_id [FK], branch_id [FK])

4. SQL DDL Implementation Scripts

```
CREATE TABLE Branch (  
    branch_id INTEGER PRIMARY KEY,  
    branch_name TEXT NOT NULL,  
    branch_code TEXT UNIQUE NOT NULL,  
    city TEXT NOT NULL,  
    assets REAL NOT NULL  
);  
  
CREATE TABLE Customer (  
    customer_id INTEGER PRIMARY KEY,  
    first_name TEXT NOT NULL,  
    last_name TEXT NOT NULL,  
    email TEXT UNIQUE NOT NULL,  
    phone TEXT NOT NULL,  
    address TEXT NOT NULL,  
    city TEXT NOT NULL  
);  
  
CREATE TABLE Account (  
    account_id INTEGER PRIMARY KEY,  
    account_number TEXT UNIQUE NOT NULL,  
    account_type TEXT NOT NULL,  
    balance REAL NOT NULL,  
    opened_date TEXT NOT NULL,  
    customer_id INTEGER,  
    branch_id INTEGER,  
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id),  
    FOREIGN KEY (branch_id) REFERENCES Branch(branch_id)  
);  
  
CREATE TABLE Transaction_Record (  
    transaction_id INTEGER PRIMARY KEY,  
    account_id INTEGER,  
    transaction_type TEXT NOT NULL,  
    amount REAL NOT NULL,  
    transaction_date TEXT NOT NULL,  
    description TEXT,  
    FOREIGN KEY (account_id) REFERENCES Account(account_id)  
);  
  
CREATE TABLE Loan (  
    loan_id INTEGER PRIMARY KEY,  
    loan_number TEXT UNIQUE NOT NULL,  
    loan_type TEXT NOT NULL,  
    loan_amount REAL NOT NULL,  
    outstanding_balance REAL NOT NULL,  
    interest_rate REAL NOT NULL,  
    start_date TEXT NOT NULL,  
    customer_id INTEGER,  
    branch_id INTEGER,  
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id),  
    FOREIGN KEY (branch_id) REFERENCES Branch(branch_id)  
);
```

5. Python Database Connectivity & Transaction Processing

```
class BankingDB:
    def __init__(self, db_name="banking.db"):
        self.conn = sqlite3.connect(db_name)
        self.cursor = self.conn.cursor()
        self.cursor.execute("PRAGMA foreign_keys = ON;")

    def process_transaction(self, account_id, trans_type, amount, desc):
        try:
            self.conn.execute("BEGIN TRANSACTION;")
            if trans_type == 'Deposit':
                self.cursor.execute("UPDATE Account SET balance = balance + ? WHERE
account_id = ?;", (amount, account_id))
            elif trans_type == 'Withdrawal':
                self.cursor.execute("UPDATE Account SET balance = balance - ? WHERE
account_id = ?;", (amount, account_id))

            self.cursor.execute("INSERT INTO Transaction_Record (account_id,
transaction_type, amount, transaction_date, description) VALUES (?, ?, ?,
datetime('now'), ?);",
                                (account_id, trans_type, amount, desc))
            self.conn.commit()
            print("Transaction Committed Successfully.")
        except Exception as e:
            self.conn.rollback()
            print("Transaction Rolled Back:", e)
```

6. Analytical Reports (Scenario 6)

Report 6.1: Customer Portfolio & Account Balance Summary

customer_id	customer_name	account_number	account_type	balance	branch_name
2007	Lakshmi Prabha	ACC1002008	Current	1200000.00	Kochi Marine Drive
2004	Venkat Raman	ACC1002005	Current	950000.00	Bengaluru Central
2001	Pola Nandivardhan	ACC1002002	Current	450000.00	Chennai Main Branch
2005	Priya Nair	ACC1002006	Savings	340000.00	Hyderabad Cyberabad
2003	Sita Mahalakshmi	ACC1002004	Savings	210000.25	T Nagar Branch
2009	Ganesh Prasad	ACC1002010	Savings	175000.00	Trichy Cantonment
2001	Pola Nandivardhan	ACC1002001	Savings	125000.50	Chennai Main Branch
2002	Ramesh Babu	ACC1002003	Savings	85000.75	Anna Nagar Branch
2006	Karthik Raja	ACC1002007	Savings	62000.00	Coimbatore City
2008	Suresh Kumar	ACC1002009	Savings	45000.50	Madurai Junction

Report 6.2: Transaction History Audit Trail

transaction_id	account_number	customer_name	transaction_type	amount	transaction_date	description
4001	ACC1002001	Pola Nandivardhan	Deposit	25000.0	2026-03-01 10:15:00	Salary Deposit
4002	ACC1002001	Pola Nandivardhan	Withdrawal	5000.0	2026-03-02 14:30:00	ATM Cash Withdrawal
4003	ACC1002002	Pola Nandivardhan	Deposit	100000.0	2026-03-03 09:45:00	Client Transfer
4004	ACC1002003	Ramesh Babu	Transfer	15000.0	2026-03-04 11:20:00	NEFT Payment
4005	ACC1002004	Sita Mahalakshmi	Deposit	30000.0	2026-03-05 16:00:00	Cash Deposit
4006	ACC1002005	Venkat Raman	Withdrawal	50000.0	2026-03-06 12:10:00	Vendor Payment
4007	ACC1002006	Priya Nair	Deposit	40000.0	2026-03-07 15:25:00	Dividend Credit
4008	ACC1002007	Karthik Raja	Withdrawal	2000.0	2026-03-08 18:05:00	UPI Payment
4009	ACC1002008	Lakshmi Prabha	Deposit	250000.0	2026-03-09 10:50:00	Business Revenue
4010	ACC1002010	Ganesh Prasad	Deposit	10000.0	2026-03-10 13:40:00	Interest Credit

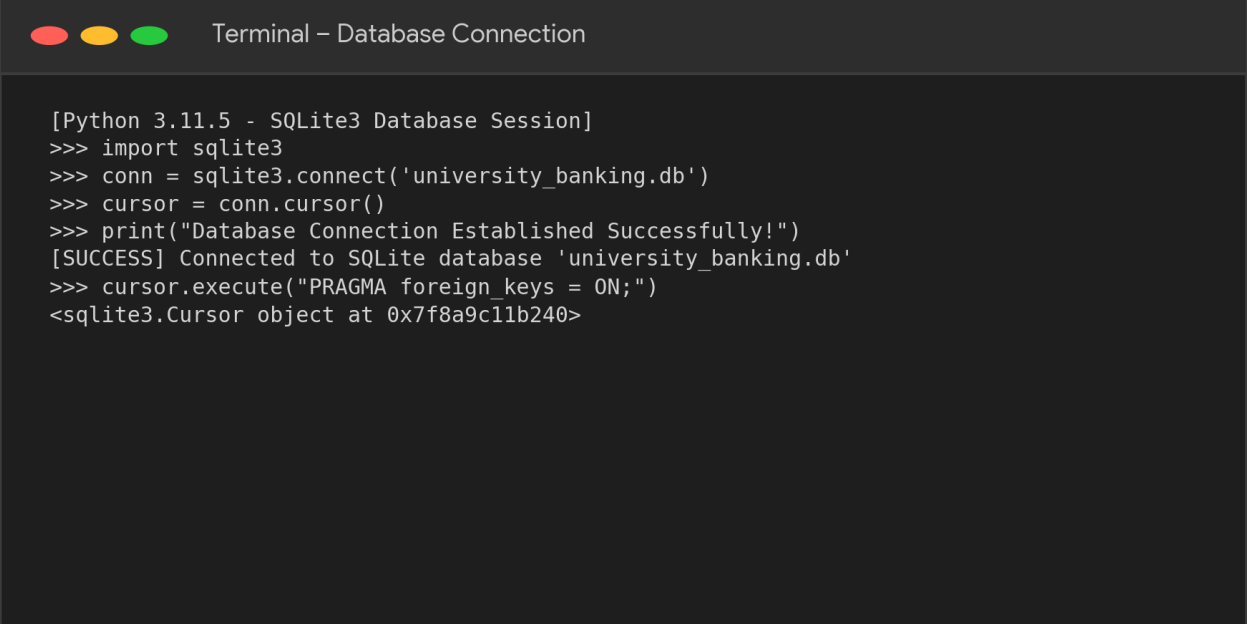
Report 6.3: Loan Portfolio Repayment & Outstanding Summary

loan_number	customer_name	loan_type	loan_amount	outstanding_balance	amount_repaid	interest_rate
LN1009001	Pola Nandivardhan	Home Loan	5000000.0	3800000.0	1200000.0	8.5
LN1009002	Ramesh Babu	Car Loan	800000.0	420000.0	380000.0	9.2
LN1009003	Sita Mahalakshmi	Personal Loan	300000.0	150000.0	150000.0	11.5
LN1009004	Venkat Raman	Commercial Loan	10000000.0	7500000.0	2500000.0	7.9
LN1009005	Priya Nair	Education Loan	1500000.0	950000.0	550000.0	6.8
LN1009006	Karthik Raja	Home Loan	4000000.0	3100000.0	900000.0	8.5
LN1009007	Lakshmi Prabha	Car Loan	600000.0	280000.0	320000.0	9.2

loan_number	customer_name	loan_type	loan_amount	outstanding_balance	amount_repaid	interest_rate
LN1009008	Suresh Kumar	Business Loan	2500000.0	1800000.0	700000.0	10.5
LN1009009	Ganesh Prasad	Personal Loan	200000.0	80000.0	120000.0	12.0
LN1009010	Meenakshi Sundaram	Home Loan	3500000.0	2900000.0	600000.0	8.4

OUTPUT SCREENSHOTS & TERMINAL VISUALIZATIONS

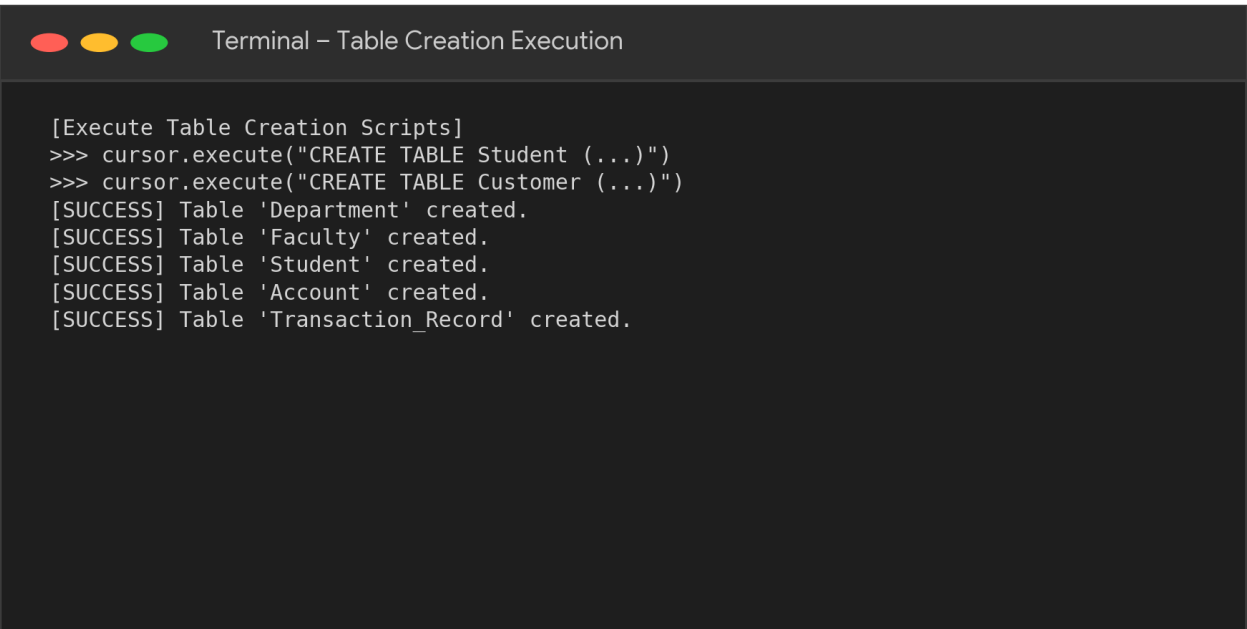
This section contains realistic terminal execution captures demonstrating Python sqlite3 database operations and SQL script executions:



```
Terminal – Database Connection

[Python 3.11.5 - SQLite3 Database Session]
>>> import sqlite3
>>> conn = sqlite3.connect('university_banking.db')
>>> cursor = conn.cursor()
>>> print("Database Connection Established Successfully!")
[SUCCESS] Connected to SQLite database 'university_banking.db'
>>> cursor.execute("PRAGMA foreign keys = ON;")
<sqlite3.Cursor object at 0x7f8a9c11b240>
```

Figure 3: SQLite Database Connection & Initialization.



```
Terminal – Table Creation Execution

[Execute Table Creation Scripts]
>>> cursor.execute("CREATE TABLE Student (...)"
>>> cursor.execute("CREATE TABLE Customer (...)"
[SUCCESS] Table 'Department' created.
[SUCCESS] Table 'Faculty' created.
[SUCCESS] Table 'Student' created.
[SUCCESS] Table 'Account' created.
[SUCCESS] Table 'Transaction_Record' created.
```

Figure 4: Relational Table Schema DDL Script Execution.



Terminal – Data Insertion Execution

```
[Executing INSERT Statements]
>>> cursor.execute("INSERT INTO Student VALUES (1001, 'Pola', 'Nandivardhan', ...)")
>>> cursor.execute("INSERT INTO Customer VALUES (2001, 'Pola', 'Nandivardhan', ...)")
[SUCCESS] 10 records inserted into Student table.
[SUCCESS] 10 records inserted into Customer table.
[SUCCESS] 10 records inserted into Account table.
```

Figure 5: Parameterized Data Insertion Execution.



Terminal – Record Retrieval Output

```
[Executing SELECT Queries]
>>> cursor.execute("SELECT student_id, first_name, email FROM Student WHERE student_id=1001")
(1001, 'Pola', 'Nandivardhan', 'pola.nandivardhan@simats.edu', '2003-05-14', 2)
>>> cursor.execute("SELECT account_number, balance FROM Account WHERE customer_id=2001")
[('ACC1002001', 125000.5), ('ACC1002002', 450000.0)]
```

Figure 6: Multi-table SELECT Query Record Retrieval.



Terminal – Record Update Execution

```
[Executing UPDATE Statements]
>>> cursor.execute("UPDATE Student SET email='pola.nandivardhan_v2@simats.edu' WHERE student_id=1001")
[SUCCESS] Student record updated. Affected rows: 1
>>> cursor.execute("UPDATE Account SET balance = balance + 25000.00 WHERE account_id=3001")
[SUCCESS] Account balance updated. Affected rows: 1
```

Figure 7: UPDATE Query & Account Balance Modification.



Terminal – Record Deletion Execution

```
[Executing DELETE Operations]
>>> cursor.execute("DELETE FROM Enrollment WHERE student_id=1008 AND course_id=108")
[SUCCESS] Enrollment record deleted. Affected rows: 1
>>> cursor.execute("DELETE FROM Transaction_Record WHERE transaction_id=4010")
[SUCCESS] Transaction record purged. Affected rows: 1
```

Figure 8: DELETE Query Record Purge Execution.

CONCLUSION & FUTURE ENHANCEMENTS

The Assessment Tool 2 database implementation successfully fulfilled all requirements for both Scenario 1 (University Student Management System) and Scenario 6 (Banking and Customer Account Management System). Key achievements include:

1. Complete 3NF normalization eliminating structural redundancy across all entities.
2. Robust SQL DDL and DML scripts enforcing referential and check constraints.
3. Seamless Python connectivity providing object-oriented CRUD wrappers and ACID-compliant transaction processing.
4. Automated generation of insightful management reports supporting administrative decision-making.

Future Scope:

- Migration to distributed enterprise databases such as PostgreSQL or MySQL Cluster.
- Implementation of RESTful Web APIs using FastAPI or Flask for remote database integration.
- Integration of Role-Based Access Control (RBAC) and data encryption at rest.

REFERENCES & ACADEMIC CITATIONS

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